SECTION 8.A

Factor Expressions

Factoring is the structural reverse of multiplication. When expressions are expanded, structure becomes hidden within combined terms. Factoring restores that structure by identifying the expressions that were multiplied together.

In this section, we focus on recognizing structure before applying procedures. Strong algebra students do not memorize isolated steps — they classify expressions by pattern. Some require a greatest common factor. Others follow trinomial structure. Some match special product patterns such as difference of squares or sum and difference of cubes.

Learning Objectives

- I can factor trinomial expressions using structural patterns including leading coefficient recognition (Level 1 and Level 2).
- I can factor expressions with a binomial greatest common factor by identifying and extracting common polynomial factors.
- I can look for and use the structure of polynomial expressions to choose and apply the most efficient factoring strategy.

Introduction

Factoring is not guesswork. It is classification. Before attempting any algebraic manipulation, determine the structure of the expression. Ask:

- Is there a greatest common factor?
- Is this a special product pattern?
- Is this a trinomial?
- Does grouping apply?

Factoring connects directly to solving equations and analyzing graphs. The factors of a polynomial determine its zeros. Those zeros determine where a graph crosses or touches the x-axis.

Real-World Connection

Engineers factor polynomial models to determine breakpoints in stress analysis. Economists factor cost and revenue equations to identify profit thresholds. Computer scientists use factoring techniques to simplify algorithms and improve computational efficiency.

Historical Note

Factoring techniques developed alongside early quadratic equation methods in ancient Babylonian mathematics and were later formalized in Islamic algebra through the work of Al-Khwarizmi. Over time, mathematicians expanded these ideas to include higher-degree polynomials and special product patterns.

Key Vocabulary and Concepts

Factor: An expression multiplied by another expression to produce a product.

Greatest Common Factor (GCF): The largest expression that divides evenly into all terms of a polynomial.

Difference of Squares:

$$a^2-b^2=(a-b)(a+b)$$

Sum of Cubes:

$$a^3+b^3=(a+b)(a^2-ab+b^2)$$

Difference of Cubes:

$$a^3-b^3=(a-b)(a^2+ab+b^2)$$

Core Strategies

Factoring should follow a logical sequence:

1. Factor out the Greatest Common Factor (GCF).
2. Look for special product patterns.
3. Factor trinomials.
4. Use grouping when necessary.
5. Factor completely.

Example 1

Factor:

$$8x^2-12x$$

Example 2

Factor:

$$x^2+11x+24$$

Example 3

Factor:

$$3x^2+10x+8$$

Example 4

Factor:

$$x^2-36$$

Example 5

Factor:

$$x^3+8$$

Example 6

Factor:

$$27x^3-64$$

Summary of Key Concepts

- Always factor out the Greatest Common Factor first.
- Recognize structural patterns before beginning algebraic manipulation.
- Difference of squares factors into two binomials.
- Sum and difference of cubes factor into a binomial and a trinomial.
- Trinomials require identifying two numbers that multiply correctly and add correctly.
- Factor completely.

Factoring Strategy Checklist

1. Rewrite the expression in descending powers.
2. Factor out the GCF.
3. Count the number of terms.
4. Check for special product patterns.
5. Apply the appropriate factoring method.
6. Verify by multiplying.

Common Mistakes

Skipping the GCF

Students often begin factoring trinomials without first removing the greatest common factor. This can make later steps more complicated and may prevent complete factoring.

Stopping Too Early

Some expressions factor in stages. For example:

$$4x^3-36x$$

First factor out the GCF, then recognize the difference of squares.

Confusing Squares and Cubes

Difference of squares applies only to expressions of the form:

$$a^2-b^2$$

Sum and difference of cubes follow different patterns.

Sign Errors in Trinomials

When factoring trinomials, students sometimes find numbers that multiply correctly but do not add correctly. Always check both multiplication and addition.

Practice Problems

Easy

1. Factor:

$$3x^2-12x$$

2. Factor:

$$x^2+7x+12$$

3. Factor:

$$x^2-25$$

4. Factor:

$$x^3+27$$

Medium

5. Factor:

$$2x^2+9x+4$$

6. Factor:

$$5x^2-20$$

7. Factor:

$$x^3-8$$

8. Factor:

$$4x^2-49$$

Hard

9. Factor completely:

$$6x^3-54x$$

10. Factor completely:

$$4x^3+16x^2-9x-36$$

11. Factor completely:

$$9x^3-72x$$

12. Factor completely:

$$x^4-16$$

Challenge

13. Create a trinomial that factors into:

$$(x+5)(x-2)$$

14. Create a difference of cubes expression whose factors include:

$$(x-4)$$

15. Create a sum of cubes expression whose factors include:

$$(x+3)$$

Justify

16. Explain why factoring must begin with identifying the greatest common factor.

17. Explain why sum and difference of cubes produce a trinomial factor.

18. Justify why factoring completely is necessary when solving polynomial equations.

Solutions

Easy

1.

$$3x^2-12x=3x(x-4)$$

2.

$$x^2+7x+12=(x+3)(x+4)$$

3.

$$x^2-25=(x-5)(x+5)$$

4.

$$x^3+27=(x+3)(x^2-3x+9)$$

Medium

5.

$$2x^2+9x+4=(2x+1)(x+4)$$

6.

$$5x^2-20=5(x^2-4)=5(x-2)(x+2)$$

7.

$$x^3-8=(x-2)(x^2+2x+4)$$

8.

$$4x^2-49=(2x-7)(2x+7)$$

Hard

9.

$$6x^3-54x=6x(x^2-9)=6x(x-3)(x+3)$$

10.

$$4x^3+16x^2-9x-36$$

Group:

$$(4x^3+16x^2)-(9x+36)$$

$$4x^2(x+4)-9(x+4)$$

$$(4x^2-9)(x+4)$$

$$(2x-3)(2x+3)(x+4)$$

11.

$$9x^3-72x=9x(x^2-8)$$

12.

$$x^4-16=(x^2-4)(x^2+4)=(x-2)(x+2)(x^2+4)$$

Challenge

13.

$$(x+5)(x-2)=x^2+3x-10$$

14. Example:

$$x^3-64=(x-4)(x^2+4x+16)$$

15. Example:

$$x^3+27=(x+3)(x^2-3x+9)$$

Justify

16. Factoring must begin with the greatest common factor because removing it simplifies the expression and reveals the remaining structure more clearly.

17. Sum and difference of cubes produce a trinomial factor because the distributive property expands into three terms that preserve the cube identity structure.

18. Factoring completely ensures that all zeros are visible. Incomplete factoring may hide additional factors necessary for solving equations.